

The inositol 1,4,5-trisphosphate receptor family

Architecture, gating, evolution and disease — a verified baseline for a genome-scale census of ITPR1/2/3

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Abstract

The inositol 1,4,5-trisphosphate receptor (IP₃R) is the endoplasmic reticulum's ligand-gated calcium-release channel and the terminal element of one of the most widely deployed signalling pathways in eukaryotic biology. Four decades after IP₃ was shown to release Ca²⁺ from an intracellular store, the receptor is among the best-characterised large channels in structural biology: a 1.3-MDa homotetramer, resolved by cryo-EM in apo, ligand-bound and lipid-embedded states, whose ligand-binding sites sit some 100 Å from the gate they control. It is also, in three of its vertebrate paralogues, a disease gene — for cerebellar ataxia, for aniridia with ataxia, for an inability to sweat, and for a demyelinating neuropathy that shades into multisystem immunodeficiency.

This review consolidates that literature with a specific downstream purpose: it is the verified baseline for a genome-scale census of the *ITPR* family, and it is therefore written to distinguish what is established from what is repeated. Every statement carries a citation to a source retrieved for this purpose; statements that could not be verified were removed rather than softened, and the four claims that failed audit are listed explicitly in §12. Two themes run through the article and matter for anything built on top of it. First, the IP₃ receptors and the ryanodine receptors are one structural superfamily, not two families that resemble each other — a fact visible in the very first sequence and confirmed domain by domain, and one that makes every sequence-similarity search for one family return the other. Second, the three vertebrate paralogues are not redundant:

they differ in ligand sensitivity, in regulation, in subcellular placement, in the organelle contacts they support, and in the diseases they cause.

Scope. Molecular architecture and structural biology (§2), ligand binding and gating (§3), regulation (§4), cellular physiology (§5), paralogue and splice diversity (§6), evolution (§7), organismal physiology from genetic models (§8), human disease genetics (§9), pharmacology and experimental tools (§10), and the questions the field cannot currently answer from its records (§11). Nomenclature: *ITPR1/2/3* for human genes, *Itpr1/2/3* for other vertebrates, IP₃R1/2/3 for proteins; InsP₃R appears in quoted titles. RyR denotes the ryanodine receptors throughout.

1. From a soluble messenger to a channel

1.1 The discovery sequence

The pathway was assembled backwards, from the message to the receiver. In 1983 IP₃ was shown to release Ca²⁺ from a non-mitochondrial intracellular store in permeabilised pancreatic acinar cells [1], and within a year the same messenger was linked to agonist-evoked Ca²⁺ mobilisation in intact cells [2]. A receptor followed: high-affinity IP₃ binding sites were characterised in brain membranes and shown to be regulated by pH and Ca²⁺ [3], and mapped across brain regions, with by far the highest density in cerebellum [4].

The protein those sites belonged to had already been purified, from a different direction entirely. P400 was purified from mouse cerebellum as a glycoprotein characteristic of Purkinje cells [5]; it was then shown to be the IP₃-binding protein itself, isolated as a large oligomeric complex [6]. Cloning in 1989 closed the loop from both ends in the same journal issue: the primary structure of P400 established it as the IP₃ receptor [7], and an independent cDNA was reported under a title that named the field's central complication at the outset — a “putative receptor for inositol 1,4,5-trisphosphate **similar to ryanodine receptor**” [8].

That the purified protein was the channel, and not an accessory to one, was settled by reconstitution: purified receptor conferred IP₃-dependent Ca²⁺ flux on reconstituted lipid vesicles [9]; an equivalent receptor was isolated and characterised from smooth muscle [10].

1.2 The first two decades

Three lines of work then developed in parallel, and their conclusions still frame the subject.

Gating is not a simple ligand switch. Single-channel recordings from cerebellar membranes revealed a bell-shaped dependence on cytosolic Ca²⁺ — activation at low concentrations, inhibition at high [11] — which made the receptor a candidate oscillator rather than a valve.

The signal is spatially organised. Ca²⁺ release proved to be built from discrete elementary events rather than a graded whole-cell response [12], organised into local and global patterns within single cells [13] and propagated between cells as intercellular waves [14].

There is more than one receptor. Cloning of additional subtypes [15] and the demonstration that they are expressed tissue- and development-specifically [16] converted a single channel into a family — one whose members would prove to differ in ligand sensitivity, regulation and cellular role (§6).

The modern consolidations of this literature are the two *Physiological Reviews* treatments [17,18], a structural-functional synthesis [19], and the historical account by one of the field's founders [20].

2. Molecular architecture

2.1 Domain organisation

Each IP₃R subunit is ~2,700 residues, of which roughly nine-tenths are cytosolic. Reading from the N-terminus, the conserved architecture is: a β -trefoil suppressor domain; the IP₃-binding core, itself a β -trefoil plus an armadillo-repeat fold; an extended central region built largely of armadillo solenoid repeats; and, at the C-terminus, a six-transmembrane pore module of the voltage-gated-channel superfamily fold, followed by a C-terminal tail that returns into the cytosolic mass.

In domain-annotation terms the diagnostic signatures are the MIR domains (PF02815) in the suppressor region, the IP₃-binding core (Ins145_P3_rec, PF08709), the RyR-IP₃R homology domain (RIH, PF01365), the RIH-associated domain (PF08454) and the Ion_trans pore domain (PF00520). Three of those five are shared with the ryanodine receptors, and the fourth — the pore — is shared with most of the cation-channel world. This is not a technicality of database annotation but a statement about descent (§7.1), and it is why domain content alone cannot assign a sequence to one family or the other.

2.2 The cryo-EM structures

Low-resolution reconstructions established the overall shape — a large cytosolic “mushroom cap” over a comparatively small membrane domain — well before the resolution revolution [21]. The near-atomic era began in 2015 with the apo structure of tetrameric rat IP₃R1 at 4.7 Å, which traced ~85% of the backbone and identified the elements involved in gating [22]. It was contemporaneous with, and interpretable alongside, the three independent near-atomic RyR1 structures published the same year [23,24,25] — the comparison that made the superfamily relationship structural rather than sequence-based.

Subsequent work has filled in the functional states. Ligand-induced allosteric rearrangements were resolved for IP₃R1 [26]; human IP₃R3 was solved in Ca²⁺- and IP₃-bound states, giving the first paralogue comparison [27]; the channel was imaged in a lipid bilayer rather than detergent [28]; and activation and gating were reconstructed across a ligand series [29], with a companion study resolving the conformational motions that couple ligand binding to the pore [30]. Structure-guided mutagenesis then identified which of the candidate Ca²⁺ sites are functionally required for activation [31]. A concise structural overview of the pre-2018 state of the field is available [32].

2.3 The pore and permeation

The pore is formed between TM5 and TM6, with a short selectivity filter and a gate at the cytosolic end of the TM6 bundle [22,27,29]. The channel is not a precision Ca²⁺ filter in the sense that voltage-gated Ca²⁺ channels are: single-channel recordings from cerebellar preparations established a large conductance with modest discrimination among divalent and monovalent cations [33]. Functionally this is appropriate — the receptor releases Ca²⁺ down a steep gradient from a store held at high concentration, so throughput matters more than selectivity.

2.4 Long-range allosteric coupling

The defining structural problem of this receptor is distance. IP₃ binds at the N-terminal core, roughly 100 Å from the gate. The cryo-EM series resolves the coupling path: ligand binding

closes the clam-shell of the binding core, that motion is transmitted through the armadillo solenoid, and the C-terminal tail — which runs from beyond TM6 back up into the cytosolic domain, forming a left-handed helical bundle at the four-fold axis and contacting the N-terminal domains of *adjacent* subunits — communicates the change to the gate [22,26,29,30].

Two consequences follow, and both recur later in this review. Because the coupling element contacts neighbouring subunits, the channel's allostery is intrinsically inter-subunit, which is the structural basis for the requirement that all four ligand sites be occupied (§3.3) and for the dominant-negative behaviour of heterozygous variants in a tetramer (§9). And because the C-terminal tail is a load-bearing mechanical element rather than a tail in the dispensable sense, variants there are not peripheral — they sit on the path that makes the channel a channel.

3. Ligand binding and gating

3.1 The IP₃-binding core and the suppressor domain

The structure of the IP₃-binding core with ligand bound [34] showed a clam-shell of two domains closing around the inositol phosphate, with the 4,5-bisphosphate pair making the specificity contacts. Apo and bound structures of the isolated ligand-binding domain confirmed the conformational cycle directly [35].

The N-terminal suppressor domain, solved separately [36], reduces IP₃ affinity — deleting it raises affinity but abolishes gating — so it is not an inhibitor but part of the transduction apparatus. Its importance is underlined from the clinical direction: a gain-of-function variant in the suppressor domain causes cerebellar ataxia [37], and the Gillespie-syndrome variants cluster in this same N-terminal region and in the channel domain (§9.1).

3.2 Ca²⁺ as obligatory co-agonist

The bell-shaped Ca²⁺ dependence [11] is the receptor's signature property, and its interpretation matured over two decades. Marchant and Taylor showed that the two ligands bind *sequentially* — IP₃ first, Ca²⁺ second — and argued that this ordering is what protects cells from spontaneous release [38]; spontaneous release does occur when the safeguard is stressed [39], and even quiescent channels show low-probability openings [40]. Systematic analysis of gating in permeabilised preparations established the co-agonist model as an allosteric scheme rather than a description [41]; the state of the activation problem before the structures arrived is reviewed in [42].

The structural counterpart arrived with the human IP₃R3 structures solved with Ca²⁺ and IP₃ [27] and was then tested functionally: of the candidate Ca²⁺-binding sites visible in the maps, mutagenesis identified which are actually required for activation [31]. The inhibitory limb of the bell — high Ca²⁺ closing the channel — remains the less well resolved half of the mechanism.

3.3 Stoichiometry: all four sites

Concatenated-subunit constructs, in which individual binding sites could be disabled within a defined tetramer, showed that Ca²⁺ release requires IP₃ occupancy at **all four** subunits [43]. This single result carries much of the interpretive weight in this review. It explains why the receptor behaves as a high-order, switch-like detector of IP₃ rather than a graded one; it explains why a mutant subunit that cannot bind or transduce ligand poisons the whole tetramer; and it is the mechanistic reason that heterozygous missense variants in *ITPR1* and *ITPR3* produce dominant disease (§9), while loss of one gene copy in isolation is comparatively well tolerated.

The question of whether the four subunits act in strict concert or with some independence is still argued, including in recent work using pore-directed peptides to disable subunits selectively.

3.4 ATP and the energy state of the cell

ATP modulates the channel at more than one site, shifting open probability without being a permissive requirement [44], and the effect differs between paralogues — the type 3 receptor's ATP regulation was characterised separately and is not identical to the type 1 [45]. Because the concentrations involved lie within the physiological range, this couples release probability to cellular energy status, and it is one of several mechanisms by which the same IP₃ signal produces different outputs in different metabolic contexts.

4. Regulation: a channel as a signalling hub

The IP₃ receptor is regulated less like an ion channel and more like a scaffold that happens to conduct. Its cytosolic mass presents a large surface for protein and post-translational input, and the collected partner list runs to dozens of proteins [46]. What follows is organised by mechanism rather than by partner, and is deliberately selective.

4.1 Phosphorylation

PKA-dependent phosphorylation was the first covalent modification described, and it *decreased* Ca²⁺ release in brain preparations [47]; protein kinase C and Ca²⁺ / calmodulin-dependent kinase were shown to phosphorylate the receptor shortly afterwards [48], and cGMP-dependent kinase adds a further site [49]. The functional sign of PKA phosphorylation is paralogue- and cell-type-dependent, which is a recurring pattern: the same modification on a different family member need not do the same thing.

Phosphorylation also acts upstream of the receptor to shape its input. Control of Ca²⁺ oscillation frequency by phosphorylation of a metabotropic glutamate receptor [50] is an early and clean demonstration that oscillation frequency is an encoded variable, not a byproduct.

4.2 The Bcl-2 family and the apoptotic set-point

The interaction between anti-apoptotic Bcl-2 proteins and the IP₃ receptor is the best-developed example of regulation with a clear pathophysiological reading. Bcl-2 binds the receptor and inhibits IP₃-evoked ER Ca²⁺ release [51], acting through its BH4 domain on the central modulatory region [52]; Mcl-1 and Bcl-2 modulate the channel in ways that protect against apoptosis [53]; and the BH4 domains of Bcl-2 and Bcl-X_L are not functionally equivalent at this target [54]. Competition experiments indicate that Bcl-2 and IP₃ can compete for the ligand-binding domain itself, modulating signalling output [55], and dual BH3-like domains have been proposed to underlie biphasic regulation of gating [56]. The area is reviewed comprehensively [57].

The therapeutic corollary is direct: peptides that disrupt the Bcl-2–IP₃R interaction reverse Bcl-2's inhibition of apoptotic Ca²⁺ signalling [58], and constitutive IP₃ signalling is what makes B-cell malignancies sensitive to the BIRD-2 disruptor [59]. The same axis is a proposed mechanism of Bcl-2's role in cancer more broadly [60].

4.3 Luminal redox and ER environment

ERp44 binds the receptor in a redox- and pH-dependent manner from the ER lumen and inhibits it, giving the channel a direct read-out of the oxidising state of the compartment it drains — and it does so subtype-specifically [61]. This is one of the clearest cases of regulation that is impossible to detect from the cytosolic side alone.

4.4 IRBIT and competitive occupancy

IRBIT binds the IP₃-binding region and is released when IP₃ rises [62]; mechanistically it suppresses receptor activity by competing with IP₃ for the same site [63]. It therefore sets a threshold: the receptor does not respond to IP₃ until the messenger has displaced a competitor, which sharpens an already high-order response (§3.3).

4.5 Calcium sensors and calmodulin

A family of neuronal Ca²⁺-binding proteins (CaBPs) was identified as protein ligands of the receptor, capable of activating it independently of IP₃ in some conditions [64], and calmodulin modulates the cardiac type 2 receptor directly [65]. Together with the intrinsic Ca²⁺ sites (§3.2), this means the channel senses Ca²⁺ through several routes at once.

4.6 Degradation and quality control

The receptor is not merely modulated but disposed of: sustained stimulation triggers ubiquitination and down-regulation [66], and an early report described Ca²⁺-induced proteolysis of the channel. The ERAD machinery involved is medically relevant in its own right — a point mutation in the ubiquitin ligase RNF170, which acts on the IP₃ receptor, causes autosomal dominant sensory ataxia [67]. That a *disposal* pathway for this receptor causes an ataxia, as loss of the receptor itself does (§9.1), is a striking convergence.

4.7 Disease proteins as regulators

Presenilins regulate IP₃R gating, and familial Alzheimer-disease presenilin mutants produce a gain-of-function enhancement of channel activity [68]. This positions the receptor downstream of a major neurodegenerative locus, and is part of the wider argument that ER Ca²⁺ handling is a convergence point in neurodegeneration [69].

5. Cellular physiology

5.1 Elementary events and the hierarchy of signals

Ca²⁺ release is quantal. Imaging in *Xenopus* oocytes resolved “puffs” — discrete, localised release events from small groups of receptors — as the elementary building blocks of the IP₃-evoked signal [12]. Below the puff sits the single-channel “blip”; above it, recruitment and regenerative coupling between release sites generate cell-wide oscillations and propagating waves [70]. Exocrine cells display the local-to-global transition particularly clearly, with agonist-evoked release beginning in a restricted apical pole before spreading [13], and waves propagate between coupled cells via IP₃ diffusion through gap junctions [14].

5.2 Oscillations: frequency as the encoded variable

That repetitive Ca²⁺ spikes are not an epiphenomenon was established early: pulsatile release does not require oscillations in IP₃ itself [71], and oscillation frequency scales with agonist concentration [72]. Two classes of model followed — one in which the oscillator is built from Ca²⁺-induced Ca²⁺ release between two pools [73], and one in which the bell-shaped Ca²⁺ dependence of a single pool of IP₃ receptors suffices [74]. The second, built directly on the gating properties in §3.2, became the standard framework. Receptor-level control of frequency was then demonstrated experimentally by manipulating upstream phosphorylation [50].

The functional significance is that the *same* second messenger can carry different instructions at different frequencies, and that the receptor's own gating kinetics are part of the encoder.

5.3 Placement: clusters, mobility and licensing

Where the receptors are matters as much as how they gate. Most IP₃ receptors are mobile and do not participate in the initial response; the Ca²⁺ signal instead initiates at a small population of *immobile* receptors positioned adjacent to ER–plasma-membrane junctions, effectively a licensed subset [75]. This resolves a long-standing discrepancy between the abundance of the protein and the sparseness of release sites, and it means receptor number is a poor proxy for release capacity — a caveat that applies directly to any inference from expression data.

5.4 Membrane contact sites: mitochondria

The receptor's best-characterised organellar partnership is with mitochondria. Close ER–mitochondria contacts determine the amplitude of the mitochondrial Ca²⁺ response [76]; the linkage is a defined structural apposition with a measurable gap and tethering elements [77]; and the resulting transfer sets the threshold for apoptosis [78]. Tethering machinery has been identified in yeast [79] and in mammalian neurons [80], and the interface is a signalling platform in its own right, hosting mTORC2–Akt signalling that feeds back on the receptor [81].

Critically for a review concerned with paralogues, **the three isoforms are not equivalent at this interface**: they differ in the ER–mitochondrial contacts they support and in the efficiency of local Ca²⁺ transfer [82]. The physiological consequences are visible at the organ level — chronic enrichment of hepatic ER–mitochondria contact drives mitochondrial Ca²⁺ overload and metabolic dysfunction in obesity [83] — and the interface is remodelled in Alzheimer disease [68].

5.5 Apoptosis and immune-cell fate

The link between the receptor and cell death was established genetically before it was structurally: T cells deficient in the IP₃ receptor resist apoptosis [84], and increased type 3 receptor expression mediates lymphocyte apoptosis [85]. The Bcl-2 axis (§4.2) supplies the regulatory logic, and the transfer geometry of §5.4 supplies the mechanism. This is also where paralogue identity becomes clinically consequential: IP₃R3 is the isoform most associated with efficient pro-apoptotic transfer, and it is the isoform whose overexpression is repeatedly reported in cancers [86,87].

5.6 Other acceptors: lysosomes

Mitochondria are not the only acceptor compartment. IP₃ receptors associate preferentially with ER–lysosome contact sites and deliver Ca²⁺ selectively to lysosomes [88], placing the receptor upstream of lysosomal Ca²⁺ handling and, by extension, of autophagic control.

6. Three paralogues, and the diversity within each

6.1 The genes

Humans carry three *ITPR* genes. Their protein products are near-identical in length — IP₃R1 2,758 aa (UniProt Q14643), IP₃R2 2,701 aa (Q14571), IP₃R3 2,671 aa (Q14573), a spread of 3% — but the genes that encode them are not built alike. Re-derived from Ensembl release 15.12 for this review:

Gene	Locus	Ensembl gene	Canonical exons	Genomic span
<i>ITPR1</i>	3p26.1	ENSG00000150995	62	354,174 bp
<i>ITPR2</i>	12p11.23	ENSG00000123104	57	497,888 bp
<i>ITPR3</i>	6p21.31	ENSG00000096433	58	76,245 bp

The genomic span varies **6.5-fold** across three genes of essentially identical exon count and protein length. *ITPR3* achieves the same architecture in roughly one-fifth of the DNA that *ITPR2* uses. Whether this reflects lineage-specific intron gain, differential loss, or selection related to the paralogues' distinct expression programmes is not established, and it is a question the genomic literature has largely not asked.

6.2 Functional differences between paralogues

The paralogues are not interchangeable. Single-channel analysis showed isoform-specific gating [89]; the type 2 receptor was identified and reconstituted separately and shown to differ functionally [90]; ATP regulation differs between types [45]; the receptors differ in IP₃ sensitivity, in Ca²⁺ dependence and in their regulation by the modulators of §4. The cleanest whole-cell demonstration remains the finding that Ca²⁺ signal *encoding* — the shape and frequency of the response — is set by which subtypes a cell expresses [91], which recasts subtype composition as a tuning parameter rather than redundancy. They also differ in the ER–mitochondrial contacts they support [82].

6.3 Heterotetramers

Because the paralogues co-assemble, most cells contain a mixed population of tetramers rather than three separate channel species. This has two consequences that matter throughout this review. It expands the functional repertoire combinatorially. And it converts a heterozygous missense variant in one gene into a defect distributed across a large fraction of a cell's channels, which — combined with the all-four-sites requirement of §3.3 — is why dominant-negative disease mechanisms are so prominent in this family (§9).

6.4 Alternative splicing

ITPR1 carries three alternatively spliced segments. SI and SII were defined in mouse, with tissue- and development-specific usage [16]; the SII region distinguishes neuronal from non-neuronal forms and the two differ in phosphorylation [92]; SII splicing was characterised in rat brain versus peripheral tissues [93]; and a third spliced segment, SIII, was identified on cloning the human cDNA [94]. Because SII lies in the regulatory region between the ligand-binding and channel domains, splicing here changes how the receptor responds rather than whether it conducts.

6.5 Tissue distribution

The classical picture — reviewed comprehensively for the three subtypes [95] — is of broad expression with strong paralogue bias:

- **IP₃R1** dominates the cerebellum, where it was discovered as the Purkinje-cell P400 antigen [5,6], and is the principal neuronal isoform [96].
- **IP₃R2** is broadly expressed, prominent in secretory epithelia [97], cardiac myocytes [98] and glia — including oligodendrocytes, where it is required for myelination [99].
- **IP₃R3** is the epithelial and secretory isoform, concentrated apically in pancreatic and salivary gland cells [100] and central to hepatobiliary function [101,87].

Two cautions. First, this distribution is not fixed: subtype expression is remodelled in disease, as shown for the two intracellular release-channel types in end-stage heart failure [102], and quantitative modern atlases have not been applied systematically across the family. Second, and more importantly, §5.3 shows that the functionally relevant quantity is not how much receptor a cell contains but how much of it is licensed and correctly placed — so expression level is a weak predictor of signalling capacity.

7. Evolution

7.1 One superfamily, not two families

The IP₃ receptors and the ryanodine receptors are a single structural superfamily. The relationship was visible in the first IP₃R sequence, whose reporting title stated it [8], and it has been confirmed at every subsequent level of resolution: the receptors share the MIR, RIH, RIH-associated and Ion_trans domains; their N-terminal regions are structurally and functionally conserved to the point that domains can be compared directly [103]; and the 2015 near-atomic structures of both families [22,23,24,25] show the same overall organisation — a vast cytosolic solenoid cap transducing ligand binding to a C-terminal pore module — at different scales (RyR subunits are ~4,900–5,000 aa, nearly twice the size).

Three consequences run through the rest of this article, and through any computational study of the family.

1. **Any similarity search for one family returns the other.** This is not a nuisance to be filtered away; it is a statement about descent, and the separation must be made by positive evidence — best-profile assignment or a labelled-bait margin — rather than by assumption. The scale of the problem is easy to underestimate: a single query for the IP₃-binding-core domain in zebrafish returns 109 protein records across 10 gene symbols, of which 53 (49%) are ryanodine receptors, including one unnamed 4,900-residue locus.
2. **RyR is the natural outgroup** for rooting an IP₃R phylogeny — better conditioned than any invertebrate IP₃R, because it is a genuine sister clade rather than a long branch within the ingroup.
3. **Size is a filter, not evidence.** The two families separate cleanly by length in well-annotated genomes, but that is a property of annotation quality in those genomes, not a phylogenetic argument.

7.2 Origins: the split predates animals

Comparative genomics places the machinery early. Analysis of genomes flanking the animal–fungal divergence — including the apusozoan *Thecamonas trahens*, from the putative unicellular sister group to Opisthokonta — finds many components of animal and fungal Ca²⁺ signalling

already present in the common ancestor, together with lineage-specific expansions of Ca²⁺ channels in the unicellular ancestors of animals and in basal fungi [104]. Homologues of both intracellular release-channel families are present in parasitic protists [105]. The broader comparative literature places Ca²⁺ signalling as an ancient and elaborate system rather than a metazoan invention [106,107], with acidic-store channels following their own evolutionary trajectory [108]; the RyR side is reviewed separately [109].

7.3 The invertebrate single-gene state

Most invertebrates carry a single *itpr*. *Drosophila* is the best-developed model: disruption of the gene affects larval metamorphosis and ecdysone release [110], genetic dissection assigns a vital requirement to aminergic neurons [111], and hypomorphs lose flight and the associated neuronal rhythmicity [112]. The single *Drosophila* receptor has been characterised biophysically and behaves recognisably like its vertebrate counterparts [113]. The *Xenopus* receptor, cloned early, similarly established conservation of structure and function across vertebrates [114].

A single-gene invertebrate state carrying out the functions distributed across three vertebrate paralogues is the strongest available argument that vertebrate paralogue specialisation is sub-functionalisation of an ancestral repertoire rather than the acquisition of new capabilities.

7.4 The vertebrate expansion — and what is not established

Vertebrates carry three paralogues; so, independently or not, do the ryanodine receptors. Teleosts add a further layer: zebrafish carries four IP₃ receptor genes — *itpr1a*, *itpr1b*, *itpr2* and *itpr3* — with the *itpr1a*/*itpr1b* pair bearing the signature of the teleost-specific whole-genome duplication (longest isoform per gene 2,635–2,819 aa, re-derived here).

Three questions about this history are **not settled by the existing literature**, and they are stated as open rather than glossed:

- Whether *ITPR1/2/3* are ohnologues from the two rounds of vertebrate whole-genome duplication has not been demonstrated with synteny-backed, phylogeny-tested evidence.
- Whether the IP₃R and RyR triplications were **independent** events is frequently asserted and, as far as this review's search could establish, nowhere demonstrated. It is an attractive parallel, not a result.
- Which two of the three vertebrate paralogues are sisters — the rooted topology of the family — is not fixed by any published, support-annotated maximum-likelihood analysis with an RyR outgroup.

7.5 The taxonomic range problem

Textbook accounts hold that land plants and fungi lack IP₃ receptors, and the model organisms support this: *Arabidopsis thaliana* and *Saccharomyces cerevisiae* carry no protein annotated with the IP₃-binding core. Yet the InterPro protein set for that same signature contains 40 Viridiplantae and 41 Fungi entries (re-derived 2026-08-18, alongside 12,149 Metazoa of 12,338 total). The lineage-specific expansion of Ca²⁺ channels reported in basal fungi [104] means a real-gene explanation cannot be dismissed a priori — but neither can mis-annotation or contamination. **Whether the famous absences are facts about genomes or facts about proteome databases is, at present, unresolved**, and it cannot be settled from database counts alone.

8. Organismal physiology: what the genetic models show

8.1 *Itpr1* — the cerebellar phenotype

Mice lacking the type 1 receptor develop ataxia and epileptic seizures and die young [115], a phenotype that matches both the receptor's Purkinje-cell enrichment [6,4] and the human ataxias caused by *ITPR1* variants (§9.1). The concordance between mouse null, spontaneous mouse mutants and human heterozygous deletion [116] is unusually tight for a large channel gene, and it is the main reason IP₃R1's cerebellar role is considered settled.

8.2 *Itpr2* and *Itpr3* — secretion, energy balance and glia

Single knockouts of *Itpr2* or *Itpr3* are comparatively mild; the double knockout is not. *Itpr2*^{-/-}*Itpr3*^{-/-} mice show severely impaired exocrine secretion — saliva, pancreatic juice, gastric acid — leading to difficulties in nutrient assimilation and an energy-metabolism phenotype [117]. This is the clearest demonstration of paralogue redundancy *within* a tissue coexisting with paralogue specialisation *between* tissues, and it matches the apical concentration of these isoforms in secretory epithelia [100].

Individually, the two genes have distinct assignments. *Itpr2* loss abolishes endothelin-1-induced arrhythmogenic Ca²⁺ signalling in atrial myocytes [98] and, in oligodendrocytes, disturbs Ca²⁺ homeostasis and impairs myelination with behavioural consequences [99]. In humans, loss of IP₃R2 function abolishes sweating [97] — a phenotype reproduced in *Itpr2*^{-/-} mice, which is the strongest single cross-species validation in the family.

8.3 Immune and haematopoietic function

T cells lacking the receptor resist apoptosis [84], and increased IP₃R3 expression mediates lymphocyte apoptosis [85]. The human genetics has since caught up with the mouse work from an unexpected direction: a recurrent dominant *ITPR3* variant produces severe immunodeficiency with CD4⁺ lymphopenia as part of a multisystem disorder [118] (§9.3).

8.4 Invertebrate physiology

The single-gene invertebrate systems (§7.3) make the pleiotropy of this receptor unusually legible: one gene, and mutants show defects in moulting and metamorphosis [110], in aminergic neuronal function [111] and in the rhythmic motor output required for flight [112]. Whatever the three vertebrate paralogues have specialised into, the ancestral gene was already doing a great deal.

9. Human disease

The *ITPR* family offers an unusually informative clinical panel: three paralogues, three largely distinct phenotypes, and dominant, dominant-negative and recessive mechanisms all represented. This makes it a test case for the proposition that pathogenic variants concentrate in the constrained parts of a protein — a proposition that can be checked against the structures of §2.

9.1 *ITPR1*: cerebellar ataxia and Gillespie syndrome

SCA15/16. Heterozygous *deletions* of *ITPR1* cause spinocerebellar ataxia type 15, identified simultaneously in human families and in the corresponding ataxic mouse [116]. The finding replicated in Japanese [119] and large European [120] cohorts, and the separately named SCA16 was shown

to be the same entity — heterozygous *ITPR1* deletion — rather than a distinct disorder [121,122]. The mechanism here is haploinsufficiency; the clinical context is reviewed among the dominant ataxias [123].

SCA29. *Missense* variants cause a congenital, non-progressive, autosomal dominant ataxia [124], consolidated as SCA29 in a multi-family case series [125]. At least one such variant is a demonstrated *gain* of function in the suppressor domain [37] — so the *ITPR1* ataxias are not a single mechanism with a single direction of effect.

Gillespie syndrome. Partial aniridia, cerebellar ataxia and intellectual disability. The genetics has two arms, and reporting only one of them is a common error: Gillespie syndrome arises from **both** biallelic recessive *ITPR1* variants **and** de novo heterozygous variants [126]. Only the latter act dominant-negatively, through a restricted repertoire of mutations concentrated in the channel domain [127] — exactly what §3.3 and §6.3 predict, since one poisoned subunit disables a tetramer that requires all four.

9.2 *ITPR2*: autosomal recessive isolated anhidrosis

Loss of IP₃R2 function causes generalised isolated anhidrosis — an inability to sweat, with morphologically normal eccrine glands [97]. Three qualifications belong with the claim. The human evidence is a *single* consanguineous family with five affected members. The variant is a homozygous *missense* in the pore-forming region that abolishes Ca²⁺ release — functionally null, not a null allele. And the strongest support is the mouse: *Itpr2*^{-/-} animals show markedly reduced sweat secretion with a corresponding Ca²⁺ defect in the gland. It is a well-supported claim resting on a narrow base, and it is the only recessive, non-neurological phenotype in the family.

9.3 *ITPR3*: neuropathy, and more than neuropathy

Dominant *ITPR3* variants cause demyelinating Charcot–Marie–Tooth disease (CMT1J). The gene–disease relationship was established with segregation and functional evidence in two families, including a variant with a dominant-negative effect on Ca²⁺ transients [128]; further families followed [129]; and a recurrent p.Thr1424Met was found in 33 affected individuals from nine unrelated families, with unusually variable age of onset and severity even within families [130]. A spontaneous canine homozygous nonsense variant reproduces the neuropathy and adds enamel defects, and reveals that loss of IP₃R3 also reduces IP₃R1 and IP₃R2 protein levels — evidence of co-regulation across the family [131].

The phenotype is **broader than neuropathy**. The recurrent de novo p.Arg2524Cys causes a complex multisystem disorder with severe immunodeficiency, CD4⁺ lymphopenia, mitochondrial dysfunction, ectodermal dysplasia and bone-marrow failure, of which CMT is one component, acting through a dominant-negative mechanism [118]. Any analysis treating the *ITPR3* variant set as purely neuropathic will mis-specify it.

9.4 Cancer

IP₃R3 is repeatedly implicated in tumour biology, with anti-apoptotic and proliferative roles reported in cancer cells [86] and increased expression enhancing malignant properties in cholangiocarcinoma [87]. The mechanistic frame is §5.4–5.5: ER-to-mitochondrial Ca²⁺ transfer sets the apoptotic threshold, and Bcl-2-family proteins tune the receptor to raise it [60]. The therapeutic reading is the BIRD-2 peptide, which kills B-cell malignancies dependent on constitutive IP₃ signalling [59].

9.5 What the variant set is good for

Because the mechanisms are heterogeneous — haploinsufficiency, gain of function, dominant-negative poisoning of a tetramer, and recessive loss — the family supports a question most disease-gene families cannot answer: *do pathogenic variants sit preferentially in the most constrained regions, and is the constrained core the same in all three paralogues?* The structural prerequisites (§2) and the clinical panel (§9.1–9.3) are both in place; what is missing is a family-wide, alignment-based constraint analysis mapped onto the cryo-EM coordinates.

10. Pharmacology and experimental tools

The IP₃ receptor has no clinically used drug and, for most of its history, no genuinely selective pharmacological tool. This is worth stating plainly, because a good deal of the literature rests on reagents whose limitations are known.

Heparin is the classical competitive antagonist at the IP₃ binding site — potent, reversible and well characterised [132] — and was used to establish IP₃ dependence of Ca²⁺ release in intact preparations [133]. It is membrane-impermeant and promiscuous (it binds many proteins), so it is a tool for permeabilised or microinjected preparations only.

2-APB is the most widely used membrane-permeant blocker and the most widely misinterpreted. It does inhibit IP₃-induced Ca²⁺ release [134], but it is not selective: it blocks store-operated Ca²⁺ entry at comparable concentrations and affects several TRP channels, and the case that it should not be used as a diagnostic for IP₃R involvement was made explicitly [135]. Results resting on 2-APB alone should be read as consistent-with rather than demonstrating IP₃R involvement.

Xestospongins were introduced as potent membrane-permeable IP₃R blockers [136] and have been valuable, though subsequent work has raised questions about selectivity and off-target actions on the ER.

Adenophostin A and its disaccharide-polyphosphate analogues are the high-potency agonists — substantially more potent than IP₃ itself — and remain the main pharmacological probes of the binding site [137].

Peptide and genetic tools have proved more selective than small molecules. BIRD-2 disrupts the Bcl-2–IP₃R interaction and has a defined mechanism and a therapeutic rationale [58,59]. Pore-directed peptides are being used to disable subunits within a tetramer, addressing the concerted-versus-independent question of §3.3. On the genetic side, the paralogue knockouts of §8 and concatenated-tetramer constructs [43] have answered questions no small molecule could.

Caged IP₃ and imaging underpin the entire elementary-event literature: photolysis of caged IP₃ with confocal or TIRF imaging is what resolved puffs [12] and what localised the licensed receptor population to ER–plasma-membrane junctions [75].

The gap is conspicuous. For a channel with three paralogues, three diseases and a documented role in tumour cell survival, there is no isoform-selective inhibitor. Structures of all three paralogues now exist (§2.2), so structure-guided design is finally tractable.

11. What the field cannot currently answer

This section is deliberately separated from the rest of the review. Everything above is supported by cited work; what follows is a list of things that are *asserted more often than they are demonstrated*, or simply unknown. They are the questions a genome-scale census of the family is designed to answer.

Q1 — Range. What is the family's true taxonomic distribution? The databases say the family is overwhelmingly metazoan (12,149 of 12,338 IP₃-binding-core proteins), yet hold 40 Viridiplantae and 41 Fungi records in a family textbooks say plants and fungi lack, while *Arabidopsis* and *S. cerevisiae* have none. Are those real genes, mis-annotations or contamination — and are the famous absences facts about genomes or about proteome databases? Nothing in the current literature settles this, and the finding of expanded Ca²⁺-channel repertoires in basal fungi [104] means the answer is not obvious.

Q2 — Origin. Are *ITPR1/2/3* ohnologues of the two rounds of vertebrate whole-genome duplication? Are teleost *itpr1a/itpr1b* products of the teleost-specific third round? Where exactly does the IP₃R/RyR split fall relative to the origin of animals? And — the claim most often repeated without support — were the IP₃R and RyR expansions to three vertebrate paralogues **independent** events? A rooted, support-annotated maximum-likelihood phylogeny with an RyR outgroup, reconciled against a calibrated species tree, would answer all four.

Q3 — Fates. Across a declared vertebrate genome scope, has any paralogue been lost, how often, and independently or once? Is any surviving as a decaying pseudogene? Loss claims in this family are currently anecdotal, and an absence claim is only as good as the assembly and annotation behind it.

Q4 — Records. How often is a real *ITPR* locus missing, fragmentary, split across gene models, unnamed, or filed under the wrong paralogue — or as a ryanodine receptor? The last risk is not hypothetical: the domain that *defines* this family returns ryanodine receptors at close to a 1:1 ratio in a routine query (§7.1), and at least one of them is an unnamed locus.

Q5 — Constraint and mechanism. Do the pathogenic variants of §9 sit in the most constrained parts of the channel, and is the constrained core the same in all three paralogues? Is the IP₃-binding core — the one module whose function RyR does not share — under different constraint from the pore, which RyR does? Does it change in lineages that have lost the upstream PLC/IP₃ pathway?

Q6 — Anything unnamed. Is there a fourth vertebrate IP₃ receptor, or an IP₃ receptor in a lineage reported to lack one?

Q7 — Gene architecture. Why does *ITPR3* achieve the same 58-exon architecture in 76 kb that *ITPR2* spreads over 498 kb (§6.1)? A 6.5-fold span difference among paralogues of identical protein length is a substantial, unexplained observation.

Q8 — Selective pharmacology. No isoform-selective inhibitor exists (§10), despite three paralogues with distinct disease associations and now-available structures for all three.

12. How this review was built, and what failed verification

12.1 Method

Candidate literature was harvested from Europe PMC across ~60 topic queries spanning the sections above, each run twice — ranked by citation count and by recency — to avoid a purely canonical or purely recent bias. Candidates were curated by hand; bibliographic metadata for every retained reference was then fetched programmatically from Europe PMC by PMID, so no author list, title, journal or year in the bibliography was transcribed manually. The final set is **137 references — 117 primary research articles and 20 reviews** — spanning 1983 to 2025, with publication type taken from the Europe PMC record rather than assigned by hand. The reference table is committed as `results/s0_baseline/references.tsv`, and this document is assembled from its section files by `scripts/s0_review_build.py`, which rennumbers the stable citation keys

into order of first appearance and renders the bibliography from that table. A cited key with no reference row is a build error, so the text and the bibliography cannot drift apart.

Database figures quoted in §6.1, §7.1, §7.5 and §11 were re-derived for this review rather than repeated: InterPro protein counts and taxonomic distributions from the InterPro API, protein lengths and domain architectures from UniProt, and gene structure from Ensembl release 15.12. The scripts are `scripts/s0_db_snapshot.py` and `scripts/s0_gene_structure.py`; the outputs are the committed tables in `results/s0_baseline/`.

Limitations. This is a narrative review, not a systematic one: there is no pre-registered protocol, no PRISMA flow, and no formal inclusion criteria beyond topical relevance and the preference for primary sources over reviews for specific claims. Search was restricted to Europe PMC and to English-language records. Coverage is deliberately deeper on structure, evolution and genetics than on cell-type-specific physiology, reflecting the downstream use.

12.2 The audit: four claims that did not survive

This review replaced a planning document whose statements were tagged by how far they could be trusted. Nineteen atomic claims were checked against the literature. Twelve were verified as written and three were kept with the wording qualified; the remaining four are recorded here because a review that silently drops what it could not confirm is less useful than one that says so.

Claim as originally written	Verdict	Resolution
Loci <i>ITPR1</i> 3p26.1, <i>ITPR2</i> 12p11.23, <i>ITPR3</i> 6p21.31	Confirmed by database	Re-derived from Ensembl; all three exact. Promoted from literature claim to database fact (§6.1).
“Each is a ~58–60 exon gene spanning hundreds of kb”	False as written	Measured: 57–62 canonical exons, and <i>ITPR3</i> spans 76 kb, not “hundreds of kb”. Replaced by the measured table in §6.1; the 6.5-fold span asymmetry became open question Q7.
“Both families independently expanded to three vertebrate paralogues”	Not established	The three-paralogue state of each family is a database fact; the <i>independence</i> of the two triplications has no primary source. Moved to open question Q2 (§7.4, §11).
Gillespie syndrome “behaves as dominant-negative in the tetramer”	Incomplete	Both biallelic recessive and de novo dominant-negative mechanisms exist [126,127]. Corrected in §9.1.

Three further claims are retained but qualified: “most agonist-evoked Ca²⁺ signals in non-muscle cells” is a review-level generalisation with no measured fraction behind it (§1.2, §5.1); the paralogue tissue-emphasis summary is qualitative and not a quantitative ranking (§6.5); and the

ITPR2 anhidrosis claim rests on a single family plus a mouse model (§9.2). One numerical correction was also applied: the tetramer is **1.3 MDa** as reported in the primary structural work [22] — four copies of the 2,758-residue human subunit give ≈ 1.25 MDa — and should not be quoted as “ ~ 1.2 MDa” unattributed.

12.3 Reading this review

Cite the sources, not this document. Where a statement here is a database figure rather than a literature claim it says so and names the release; those figures should be re-derived before being quoted, because they will move.

References

1. Streb H, Irvine RF, Berridge MJ, Schulz I. Release of Ca²⁺ from a nonmitochondrial intracellular store in pancreatic acinar cells by inositol-1,4,5-trisphosphate. *Nature* **1983**. PMID [6605482](#). doi:[10.1038/306067a0](#).
2. Berridge MJ, Heslop JP, Irvine RF, Brown KD. Inositol trisphosphate formation and calcium mobilization in Swiss 3T3 cells in response to platelet-derived growth factor. *Biochem J* **1984**. PMID [6089758](#). doi:[10.1042/bj2220195](#).
3. Worley PF, Baraban JM, Supattapone S, Wilson VS, Snyder SH. Characterization of inositol trisphosphate receptor binding in brain. Regulation by pH and calcium. *J Biol Chem* **1987**. PMID [3040730](#). doi:[10.1016/s0021-9258\(18\)45326-4](#).
4. Worley PF, Baraban JM, Colvin JS, Snyder SH. Inositol trisphosphate receptor localization in brain: variable stoichiometry with protein kinase C. *Nature* **1987**. PMID [3027583](#). doi:[10.1038/325159a0](#).
5. Maeda N, Niinobe M, Nakahira K, Mikoshiba K. Purification and characterization of P400 protein, a glycoprotein characteristic of Purkinje cell, from mouse cerebellum. *J Neurochem* **1988**. PMID [3141586](#). doi:[10.1111/j.1471-4159.1988.tb01151.x](#).
6. Maeda N, Niinobe M, Mikoshiba K. A cerebellar Purkinje cell marker P400 protein is an inositol 1,4,5-trisphosphate (InsP₃) receptor protein. Purification and characterization of InsP₃ receptor complex. *EMBO J* **1990**. PMID [2153079](#). doi:[10.1002/j.1460-2075.1990.tb07386.x](#).
7. Furuichi T, Yoshikawa S, Miyawaki A, Wada K, Maeda N, Mikoshiba K. Primary structure and functional expression of the inositol 1,4,5-trisphosphate-binding protein P400. *Nature* **1989**. PMID [2554142](#). doi:[10.1038/342032a0](#).
8. Mignery GA, Sudhof TC, Takei K, De Camilli P. Putative receptor for inositol 1,4,5-trisphosphate similar to ryanodine receptor. *Nature* **1989**. PMID [2554146](#). doi:[10.1038/342192a0](#).
9. Ferris CD, Huganir RL, Supattapone S, Snyder SH. Purified inositol 1,4,5-trisphosphate receptor mediates calcium flux in reconstituted lipid vesicles. *Nature* **1989**. PMID [2554143](#). doi:[10.1038/342087a0](#).
10. Chadwick CC, Saito A, Fleischer S. Isolation and characterization of the inositol trisphosphate receptor from smooth muscle. *Proc Natl Acad Sci USA* **1990**. PMID [2156261](#). doi:[10.1073/pnas.87.6.2132](#).
11. Bezprozvanny I, Watras J, Ehrlich BE. Bell-shaped calcium-response curves of Ins(1,4,5)P₃- and calcium-gated channels from endoplasmic reticulum of cerebellum. *Nature* **1991**. PMID [1648178](#). doi:[10.1038/351751a0](#).
12. Yao Y, Choi J, Parker I. Quantal puffs of intracellular Ca²⁺ evoked by inositol trisphosphate in *Xenopus* oocytes. *J Physiol* **1995**. PMID [7738847](#). doi:[10.1113/jphysiol.1995.sp020703](#).

13. Thorn P, Lawrie AM, Smith PM, Gallacher DV, Petersen OH. Local and global cytosolic Ca²⁺ oscillations in exocrine cells evoked by agonists and inositol trisphosphate. *Cell* **1993**. PMID [8395347](#). doi:[10.1016/0092-8674\(93\)90513-p](#).
14. Boitano S, Dirksen ER, Sanderson MJ. Intercellular propagation of calcium waves mediated by inositol trisphosphate. *Science* **1992**. PMID [1411526](#). doi:[10.1126/science.1411526](#).
15. Ross CA, Danoff SK, Schell MJ, Snyder SH, Ullrich A. Three additional inositol 1,4,5-trisphosphate receptors: molecular cloning and differential localization in brain and peripheral tissues. *Proc Natl Acad Sci USA* **1992**. PMID [1374893](#). doi:[10.1073/pnas.89.10.4265](#).
16. Nakagawa T, Okano H, Furuichi T, Aruga J, Mikoshiba K. The subtypes of the mouse inositol 1,4,5-trisphosphate receptor are expressed in a tissue-specific and developmentally specific manner. *Proc Natl Acad Sci USA* **1991**. PMID [1648733](#). doi:[10.1073/pnas.88.14.6244](#).
17. Foskett JK, White C, Cheung KH, Mak DO. Inositol trisphosphate receptor Ca²⁺ release channels. *Physiol Rev* **2007**. PMID [17429043](#). doi:[10.1152/physrev.00035.2006](#).
18. Berridge MJ. The Inositol Trisphosphate/Calcium Signaling Pathway in Health and Disease. *Physiol Rev* **2016**. PMID [27512009](#). doi:[10.1152/physrev.00006.2016](#).
19. Prole DL, Taylor CW. Structure and Function of IP₃ Receptors. *Cold Spring Harb Perspect Biol* **2019**. PMID [30745293](#). doi:[10.1101/cshperspect.a035063](#).
20. Mikoshiba K. IP₃ receptor/Ca²⁺ channel: from discovery to new signaling concepts. *J Neurochem* **2007**. PMID [17697045](#). doi:[10.1111/j.1471-4159.2007.04825.x](#).
21. Serysheva II, Bare DJ, Ludtke SJ, Kettlun CS, Chiu W, Mignery GA. Structure of the type 1 inositol 1,4,5-trisphosphate receptor revealed by electron cryomicroscopy. *J Biol Chem* **2003**. PMID [12714606](#). doi:[10.1074/jbc.c300148200](#).
22. Fan G, Baker ML, Wang Z, Baker MR, Sinyagovskiy PA, Chiu W *et al*. Gating machinery of InsP₃R channels revealed by electron cryomicroscopy. *Nature* **2015**. PMID [26458101](#). doi:[10.1038/nature15249](#).
23. Zalk R, Clarke OB, des Georges A, Grassucci RA, Reiken S, Mancina F *et al*. Structure of a mammalian ryanodine receptor. *Nature* **2015**. PMID [25470061](#). doi:[10.1038/nature13950](#).
24. Yan Z, Bai X, Yan C, Wu J, Li Z, Xie T *et al*. Structure of the rabbit ryanodine receptor RyR1 at near-atomic resolution. *Nature* **2015**. PMID [25517095](#). doi:[10.1038/nature14063](#).
25. Efremov RG, Leitner A, Aebersold R, Raunser S. Architecture and conformational switch mechanism of the ryanodine receptor. *Nature* **2015**. PMID [25470059](#). doi:[10.1038/nature13916](#).
26. Fan G, Baker MR, Wang Z, Seryshev AB, Ludtke SJ, Baker ML *et al*. Cryo-EM reveals ligand and induced allostery underlying InsP₃R channel gating. *Cell Res* **2018**. PMID [30470765](#). doi:[10.1038/s41422-018-0108-5](#).
27. Paknejad N, Hite RK. Structural basis for the regulation of inositol trisphosphate receptors by Ca²⁺ and IP₃. *Nat Struct Mol Biol* **2018**. PMID [30013099](#). doi:[10.1038/s41594-018-0089-6](#).
28. Baker MR, Fan G, Seryshev AB, Agosto MA, Baker ML, Serysheva II. Cryo-EM structure of type 1 IP₃R channel in a lipid bilayer. *Commun Biol* **2021**. PMID [34035440](#). doi:[10.1038/s42003-021-02156-4](#).
29. Schmitz EA, Takahashi H, Karakas E. Structural basis for activation and gating of IP₃ receptors. *Nat Commun* **2022**. PMID [35301323](#). doi:[10.1038/s41467-022-29073-2](#).
30. Fan G, Baker MR, Terry LE, Arige V, Chen M, Seryshev AB *et al*. Conformational motions and ligand-binding underlying gating and regulation in IP₃R channel. *Nat Commun* **2022**. PMID [36376291](#). doi:[10.1038/s41467-022-34574-1](#).
31. Arige V, Terry LE, Wagner LE, Malik S, Baker MR, Fan G *et al*. Functional determination of calcium-binding sites required for the activation of inositol 1,4,5-trisphosphate receptors. *Proc Natl Acad Sci USA* **2022**. PMID [36122240](#). doi:[10.1073/pnas.2209267119](#).
32. Baker MR, Fan G, Serysheva II. Structure of IP₃R channel: high-resolution insights from

- cryo-EM. *Curr Opin Struct Biol* **2017**. PMID [28618351](#). doi:[10.1016/j.sbi.2017.05.014](#).
33. Bezprozvanny I, Ehrlich BE. Inositol (1,4,5)-trisphosphate (InsP₃)-gated Ca channels from cerebellum: conduction properties for divalent cations and regulation by intraluminal calcium. *J Gen Physiol* **1994**. PMID [7876825](#). doi:[10.1085/jgp.104.5.821](#).
34. Bosanac I, Alattia JR, Mal TK, Chan J, Talarico S, Tong FK *et al*. Structure of the inositol 1,4,5-trisphosphate receptor binding core in complex with its ligand. *Nature* **2002**. PMID [12442173](#). doi:[10.1038/nature01268](#).
35. Lin CC, Baek K, Lu Z. Apo and InsP₃-bound crystal structures of the ligand-binding domain of an InsP₃ receptor. *Nat Struct Mol Biol* **2011**. PMID [21892169](#). doi:[10.1038/nsmb.2112](#).
36. Bosanac I, Yamazaki H, Matsu-Ura T, Michikawa T, Mikoshiba K, Ikura M. Crystal structure of the ligand binding suppressor domain of type 1 inositol 1,4,5-trisphosphate receptor. *Mol Cell* **2005**. PMID [15664189](#). doi:[10.1016/j.molcel.2004.11.046](#).
37. Casey JP, Hirouchi T, Hisatsune C, Lynch B, Murphy R, Dunne AM *et al*. A novel gain-of-function mutation in the ITPR1 suppressor domain causes spinocerebellar ataxia with altered Ca²⁺ signal patterns. *J Neurol* **2017**. PMID [28620721](#). doi:[10.1007/s00415-017-8545-5](#).
38. Marchant JS, Taylor CW. Cooperative activation of IP₃ receptors by sequential binding of IP₃ and Ca²⁺ safeguards against spontaneous activity. *Curr Biol* **1997**. PMID [9210378](#). doi:[10.1016/s0960-9822\(06\)00230-5](#).
39. Missiaen L, Taylor CW, Berridge MJ. Spontaneous calcium release from inositol trisphosphate-sensitive calcium stores. *Nature* **1991**. PMID [1857419](#). doi:[10.1038/352241a0](#).
40. Mak DO, McBride SM, Foskett JK. Spontaneous channel activity of the inositol 1,4,5-trisphosphate (InsP₃) receptor (InsP₃R). Application of allosteric modeling to calcium and InsP₃ regulation of InsP₃R single-channel gating. *J Gen Physiol* **2003**. PMID [14581584](#). doi:[10.1085/jgp.200308809](#).
41. Hirose K, Kadowaki S, Iino M. Allosteric regulation by cytoplasmic Ca²⁺ and IP₃ of the gating of IP₃ receptors in permeabilized guinea-pig vascular smooth muscle cells. *J Physiol* **1998**. PMID [9490868](#). doi:[10.1111/j.1469-7793.1998.407bw.x](#).
42. Taylor CW, Tovey SC. IP(3) receptors: toward understanding their activation. *Cold Spring Harb Perspect Biol* **2010**. PMID [20980441](#). doi:[10.1101/cshperspect.a004010](#).
43. Alzayady KJ, Wang L, Chandrasekhar R, Wagner LE, Van Petegem F, Yule DI. Defining the stoichiometry of inositol 1,4,5-trisphosphate binding required to initiate Ca²⁺ release. *Sci Signal* **2016**. PMID [27048566](#). doi:[10.1126/scisignal.aad6281](#).
44. Bezprozvanny I, Ehrlich BE. ATP modulates the function of inositol 1,4,5-trisphosphate-gated channels at two sites. *Neuron* **1993**. PMID [7686381](#). doi:[10.1016/0896-6273\(93\)90319-m](#).
45. Mak DO, McBride S, Foskett JK. ATP regulation of recombinant type 3 inositol 1,4,5-trisphosphate receptor gating. *J Gen Physiol* **2001**. PMID [11331355](#). doi:[10.1085/jgp.117.5.447](#).
46. Prole DL, Taylor CW. Inositol 1,4,5-trisphosphate receptors and their protein partners as signalling hubs. *J Physiol* **2016**. PMID [26830355](#). doi:[10.1113/jp271139](#).
47. Supattapone S, Danoff SK, Theibert A, Joseph SK, Steiner J, Snyder SH. Cyclic AMP-dependent phosphorylation of a brain inositol trisphosphate receptor decreases its release of calcium. *Proc Natl Acad Sci USA* **1988**. PMID [2847175](#). doi:[10.1073/pnas.85.22.8747](#).
48. Ferris CD, Huganir RL, Bredt DS, Cameron AM, Snyder SH. Inositol trisphosphate receptor: phosphorylation by protein kinase C and calcium calmodulin-dependent protein kinases in reconstituted lipid vesicles. *Proc Natl Acad Sci USA* **1991**. PMID [1848697](#). doi:[10.1073/pnas.88.6.2232](#).
49. Komalavilas P, Lincoln TM. Phosphorylation of the inositol 1,4,5-trisphosphate receptor by

- cyclic GMP-dependent protein kinase. *J Biol Chem* **1994**. PMID [8132598](#). doi:[10.1016/s0021-9258\(17\)37024-2](#).
50. Kawabata S, Tsutsumi R, Kohara A, Yamaguchi T, Nakanishi S, Okada M. Control of calcium oscillations by phosphorylation of metabotropic glutamate receptors. *Nature* **1996**. PMID [8779726](#). doi:[10.1038/383089a0](#).
51. Chen R, Valencia I, Zhong F, McColl KS, Roderick HL, Bootman MD *et al*. Bcl-2 functionally interacts with inositol 1,4,5-trisphosphate receptors to regulate calcium release from the ER in response to inositol 1,4,5-trisphosphate. *J Cell Biol* **2004**. PMID [15263017](#). doi:[10.1083/jcb.200402193](#).
52. Rong YP, Bultynck G, Aromolaran AS, Zhong F, Parys JB, De Smedt H *et al*. The BH4 domain of Bcl-2 inhibits ER calcium release and apoptosis by binding the regulatory and coupling domain of the IP₃ receptor. *Proc Natl Acad Sci USA* **2009**. PMID [19706527](#). doi:[10.1073/pnas.0907555106](#).
53. Eckenrode EF, Yang J, Velmurugan GV, Foskett JK, White C. Apoptosis protection by Mcl-1 and Bcl-2 modulation of inositol 1,4,5-trisphosphate receptor-dependent Ca²⁺ signaling. *J Biol Chem* **2010**. PMID [20189983](#). doi:[10.1074/jbc.m109.096040](#).
54. Monaco G, Decrock E, Akl H, Ponsaerts R, Vervliet T, Luyten T *et al*. Selective regulation of IP₃-receptor-mediated Ca²⁺ signaling and apoptosis by the BH4 domain of Bcl-2 versus Bcl-XL. *Cell Death Differ* **2012**. PMID [21818117](#). doi:[10.1038/cdd.2011.97](#).
55. Ivanova H, Wagner LE, Tanimura A, Vandermarliere E, Luyten T, Welkenhuyzen K *et al*. Bcl-2 and IP₃ compete for the ligand-binding domain of IP₃Rs modulating Ca²⁺ signaling output. *Cell Mol Life Sci* **2019**. PMID [30989245](#). doi:[10.1007/s00018-019-03091-8](#).
56. Yang J, Vais H, Gu W, Foskett JK. Biphasic regulation of InsP₃ receptor gating by dual Ca²⁺ release channel BH3-like domains mediates Bcl-xL control of cell viability. *Proc Natl Acad Sci USA* **2016**. PMID [26976600](#). doi:[10.1073/pnas.1517935113](#).
57. Ivanova H, Vervliet T, Monaco G, Terry LE, Rosa N, Baker MR *et al*. Bcl-2-Protein Family as Modulators of IP₃ Receptors and Other Organellar Ca²⁺ Channels. *Cold Spring Harb Perspect Biol* **2020**. PMID [31501195](#). doi:[10.1101/cshperspect.a035089](#).
58. Rong YP, Aromolaran AS, Bultynck G, Zhong F, Li X, McColl K *et al*. Targeting Bcl-2-IP₃ receptor interaction to reverse Bcl-2's inhibition of apoptotic calcium signals. *Mol Cell* **2008**. PMID [18657507](#). doi:[10.1016/j.molcel.2008.06.014](#).
59. Bittremieux M, La Rovere RM, Akl H, Martinez C, Welkenhuyzen K, Dubron K *et al*. Constitutive IP₃ signaling underlies the sensitivity of B-cell cancers to the Bcl-2/IP₃ receptor disruptor BIRD-2. *Cell Death Differ* **2019**. PMID [29899382](#). doi:[10.1038/s41418-018-0142-3](#).
60. Akl H, Vervloessem T, Kiviluoto S, Bittremieux M, Parys JB, De Smedt H *et al*. A dual role for the anti-apoptotic Bcl-2 protein in cancer: mitochondria versus endoplasmic reticulum. *Biochim Biophys Acta* **2014**. PMID [24768714](#). doi:[10.1016/j.bbamcr.2014.04.017](#).
61. Higo T, Hattori M, Nakamura T, Natsume T, Michikawa T, Mikoshiba K. Subtype-specific and ER lumenal environment-dependent regulation of inositol 1,4,5-trisphosphate receptor type 1 by ERp44. *Cell* **2005**. PMID [15652484](#). doi:[10.1016/j.cell.2004.11.048](#).
62. Ando H, Mizutani A, Matsu-ura T, Mikoshiba K. IRBIT, a novel inositol 1,4,5-trisphosphate (IP₃) receptor-binding protein, is released from the IP₃ receptor upon IP₃ binding to the receptor. *J Biol Chem* **2003**. PMID [12525476](#). doi:[10.1074/jbc.M210119200](#).
63. Ando H, Mizutani A, Kiefer H, Tsuzurugi D, Michikawa T, Mikoshiba K. IRBIT suppresses IP₃ receptor activity by competing with IP₃ for the common binding site on the IP₃ receptor. *Mol Cell* **2006**. PMID [16793548](#). doi:[10.1016/j.molcel.2006.05.017](#).
64. Yang J, McBride S, Mak DO, Vardi N, Palczewski K, Haeseleer F *et al*. Identification of a family of calcium sensors as protein ligands of inositol trisphosphate receptor Ca(2+) release

- channels. *Proc Natl Acad Sci USA* **2002**. PMID [12032348](#). doi:[10.1073/pnas.102006299](#).
65. Bare DJ, Kettlun CS, Liang M, Bers DM, Mignery GA. Cardiac type 2 inositol 1,4,5-trisphosphate receptor: interaction and modulation by calcium/calmodulin-dependent protein kinase II. *J Biol Chem* **2005**. PMID [15710625](#). doi:[10.1074/jbc.m414212200](#).
66. Wojcikiewicz RJ, Ernst SA, Yule DI. Secretagogues cause ubiquitination and down-regulation of inositol 1, 4,5-trisphosphate receptors in rat pancreatic acinar cells. *Gastroenterology* **1999**. PMID [10220512](#). doi:[10.1016/s0016-5085\(99\)70023-5](#).
67. Wright FA, Lu JP, Sliter DA, Dupré N, Rouleau GA, Wojcikiewicz RJ. A Point Mutation in the Ubiquitin Ligase RNF170 That Causes Autosomal Dominant Sensory Ataxia Destabilizes the Protein and Impairs Inositol 1,4,5-Trisphosphate Receptor-mediated Ca²⁺ Signaling. *J Biol Chem* **2015**. PMID [25882839](#). doi:[10.1074/jbc.m115.655043](#).
68. Cheung KH, Shineman D, Müller M, Cárdenas C, Mei L, Yang J *et al.* Mechanism of Ca²⁺ disruption in Alzheimer's disease by presenilin regulation of InsP₃ receptor channel gating. *Neuron* **2008**. PMID [18579078](#). doi:[10.1016/j.neuron.2008.04.015](#).
69. Foskett JK. Inositol trisphosphate receptor Ca²⁺ release channels in neurological diseases. *Pflugers Arch* **2010**. PMID [20383523](#). doi:[10.1007/s00424-010-0826-0](#).
70. Berridge MJ. Inositol trisphosphate and calcium signalling. *Nature* **1993**. PMID [8381210](#). doi:[10.1038/361315a0](#).
71. Wakui M, Potter BV, Petersen OH. Pulsatile intracellular calcium release does not depend on fluctuations in inositol trisphosphate concentration. *Nature* **1989**. PMID [2498663](#). doi:[10.1038/339317a0](#).
72. Harootunian AT, Kao JP, Paranjape S, Tsien RY. Generation of calcium oscillations in fibroblasts by positive feedback between calcium and IP₃. *Science* **1991**. PMID [1986413](#). doi:[10.1126/science.1986413](#).
73. Goldbeter A, Dupont G, Berridge MJ. Minimal model for signal-induced Ca²⁺ oscillations and for their frequency encoding through protein phosphorylation. *Proc Natl Acad Sci USA* **1990**. PMID [2304911](#). doi:[10.1073/pnas.87.4.1461](#).
74. De Young GW, Keizer J. A single-pool inositol 1,4,5-trisphosphate-receptor-based model for agonist-stimulated oscillations in Ca²⁺ concentration. *Proc Natl Acad Sci USA* **1992**. PMID [1329108](#). doi:[10.1073/pnas.89.20.9895](#).
75. Thillaiappan NB, Chavda AP, Tovey SC, Prole DL, Taylor CW. Ca²⁺ signals initiate at immobile IP₃ receptors adjacent to ER-plasma membrane junctions. *Nat Commun* **2017**. PMID [29138405](#). doi:[10.1038/s41467-017-01644-8](#).
76. Rizzuto R, Pinton P, Carrington W, Fay FS, Fogarty KE, Lifshitz LM *et al.* Close contacts with the endoplasmic reticulum as determinants of mitochondrial Ca²⁺ responses. *Science* **1998**. PMID [9624056](#). doi:[10.1126/science.280.5370.1763](#).
77. Csordas G, Renken C, Varnai P, Walter L, Weaver D, Buttle KF *et al.* Structural and functional features and significance of the physical linkage between ER and mitochondria. *J Cell Biol* **2006**. PMID [16982799](#). doi:[10.1083/jcb.200604016](#).
78. Pinton P, Giorgi C, Siviero R, Zecchini E, Rizzuto R. Calcium and apoptosis: ER-mitochondria Ca²⁺ transfer in the control of apoptosis. *Oncogene* **2008**. PMID [18955969](#). doi:[10.1038/onc.2008.308](#).
79. Kornmann B, Currie E, Collins SR, Schuldiner M, Nunnari J, Weissman JS *et al.* An ER-mitochondria tethering complex revealed by a synthetic biology screen. *Science* **2009**. PMID [19556461](#). doi:[10.1126/science.1175088](#).
80. Hirabayashi Y, Kwon SK, Paek H, Pernice WM, Paul MA, Lee J *et al.* ER-mitochondria tethering by PDZD8 regulates Ca²⁺ dynamics in mammalian neurons. *Science* **2017**. PMID [29097544](#). doi:[10.1126/science.aan6009](#).

81. Betz C, Stracka D, Prescianotto-Baschong C, Frieden M, Demaurex N, Hall MN. Feature Article: mTOR complex 2-Akt signaling at mitochondria-associated endoplasmic reticulum membranes (MAM) regulates mitochondrial physiology. *Proc Natl Acad Sci USA* **2013**. PMID [23852728](#). doi:[10.1073/pnas.1302455110](#).
82. Bartok A, Weaver D, Golenar T, Nichtova Z, Katona M, Bansaghi S *et al*. IP₃ receptor isoforms differently regulate ER-mitochondrial contacts and local calcium transfer. *Nat Commun* **2019**. PMID [31427578](#). doi:[10.1038/s41467-019-11646-3](#).
83. Arruda AP, Pers BM, Parlakg  l G, G  ney E, Inouye K, Hotamisligil GS. Chronic enrichment of hepatic endoplasmic reticulum-mitochondria contact leads to mitochondrial dysfunction in obesity. *Nat Med* **2014**. PMID [25419710](#). doi:[10.1038/nm.3735](#).
84. Jayaraman T, Marks AR. T cells deficient in inositol 1,4,5-trisphosphate receptor are resistant to apoptosis. *Mol Cell Biol* **1997**. PMID [9154798](#). doi:[10.1128/mcb.17.6.3005](#).
85. Khan AA, Soloski MJ, Sharp AH, Schilling G, Sabatini DM, Li SH *et al*. Lymphocyte apoptosis: mediation by increased type 3 inositol 1,4,5-trisphosphate receptor. *Science* **1996**. PMID [8662540](#). doi:[10.1126/science.273.5274.503](#).
86. Rezuchova I, Hudecova S, Soltysova A, Matuskova M, Durinikova E, Chovancova B *et al*. Type 3 inositol 1,4,5-trisphosphate receptor has antiapoptotic and proliferative role in cancer cells. *Cell Death Dis* **2019**. PMID [30796197](#). doi:[10.1038/s41419-019-1433-4](#).
87. Ueasilamongkol P, Khamphaya T, Guerra MT, Rodrigues MA, Gomes DA, Kong Y *et al*. Type 3 Inositol 1,4,5-Trisphosphate Receptor Is Increased and Enhances Malignant Properties in Cholangiocarcinoma. *Hepatology* **2020**. PMID [31251815](#). doi:[10.1002/hep.30839](#).
88. Atakpa P, Thillaiappan NB, Mataragka S, Prole DL, Taylor CW. IP₃ Receptors Preferentially Associate with ER-Lysosome Contact Sites and Selectively Deliver Ca²⁺ to Lysosomes. *Cell Rep* **2018**. PMID [30540949](#). doi:[10.1016/j.celrep.2018.11.064](#).
89. Ramos-Franco J, Fill M, Mignery GA. Isoform-specific function of single inositol 1,4,5-trisphosphate receptor channels. *Biophys J* **1998**. PMID [9675184](#). doi:[10.1016/s0006-3495\(98\)77572-1](#).
90. Perez PJ, Ramos-Franco J, Fill M, Mignery GA. Identification and functional reconstitution of the type 2 inositol 1,4,5-trisphosphate receptor from ventricular cardiac myocytes. *J Biol Chem* **1997**. PMID [9295347](#). doi:[10.1074/jbc.272.38.23961](#).
91. Miyakawa T, Maeda A, Yamazawa T, Hirose K, Kurosaki T, Iino M. Encoding of Ca²⁺ signals by differential expression of IP₃ receptor subtypes. *EMBO J* **1999**. PMID [10064596](#). doi:[10.1093/emboj/18.5.1303](#).
92. Danoff SK, Ferris CD, Donath C, Fischer GA, Munemitsu S, Ullrich A *et al*. Inositol 1,4,5-trisphosphate receptors: distinct neuronal and nonneuronal forms derived by alternative splicing differ in phosphorylation. *Proc Natl Acad Sci USA* **1991**. PMID [1849282](#). doi:[10.1073/pnas.88.7.2951](#).
93. Schell MJ, Danoff SK, Ross CA. Inositol (1,4,5)-trisphosphate receptor: characterization of neuron-specific alternative splicing in rat brain and peripheral tissues. *Brain Res Mol Brain Res* **1993**. PMID [8389956](#). doi:[10.1016/0169-328x\(93\)90004-9](#).
94. Nucifora FC, Li SH, Danoff S, Ullrich A, Ross CA. Molecular cloning of a cDNA for the human inositol 1,4,5-trisphosphate receptor type 1, and the identification of a third alternatively spliced variant. *Brain Res Mol Brain Res* **1995**. PMID [7500840](#). doi:[10.1016/0169-328x\(95\)00194-w](#).
95. Taylor CW, Genazzani AA, Morris SA. Expression of inositol trisphosphate receptors. *Cell Calcium* **1999**. PMID [10668562](#). doi:[10.1054/ceca.1999.0034](#).
96. Sharp AH, Nucifora FC, Blondel O, Sheppard CA, Zhang C, Snyder SH *et al*. Differential cellular expression of isoforms of inositol 1,4,5-trisphosphate receptors in

- neurons and glia in brain. *J Comp Neurol* **1999**. PMID [10096607](#). doi:[10.1002/\(sici\)1096-9861\(19990405\)406:23.0.co;2-7](#).
97. Klar J, Hisatsune C, Baig SM, Tariq M, Johansson AC, Rasool M *et al*. Abolished InsP3R2 function inhibits sweat secretion in both humans and mice. *J Clin Invest* **2014**. PMID [25329695](#). doi:[10.1172/JCI78173](#).
98. Li X, Zima AV, Sheikh F, Blatter LA, Chen J. Endothelin-1-induced arrhythmogenic Ca²⁺ signaling is abolished in atrial myocytes of inositol-1,4,5-trisphosphate(IP3)-receptor type 2-deficient mice. *Circ Res* **2005**. PMID [15933266](#). doi:[10.1161/01.res.0000172556.05576.4c](#).
99. Zhang M, Zhi N, Feng J, Liu Y, Zhang M, Liu D *et al*. ITPR2 Mediated Calcium Homeostasis in Oligodendrocytes is Essential for Myelination and Involved in Depressive-Like Behavior in Adolescent Mice. *Adv Sci* **2024**. PMID [38476116](#). doi:[10.1002/advs.202306498](#).
100. Lee MG, Xu X, Zeng W, Diaz J, Wojcikiewicz RJ, Kuo TH *et al*. Polarized expression of Ca²⁺ channels in pancreatic and salivary gland cells. Correlation with initiation and propagation of [Ca²⁺]_i waves. *J Biol Chem* **1997**. PMID [9188472](#). doi:[10.1074/jbc.272.25.15765](#).
101. Mangla A, Guerra MT, Nathanson MH. Type 3 inositol 1,4,5-trisphosphate receptor: A calcium channel for all seasons. *Cell Calcium* **2020**. PMID [31790953](#). doi:[10.1016/j.ceca.2019.102132](#).
102. Go LO, Moschella MC, Watras J, Handa KK, Fyfe BS, Marks AR. Differential regulation of two types of intracellular calcium release channels during end-stage heart failure. *J Clin Invest* **1995**. PMID [7860772](#). doi:[10.1172/jci117739](#).
103. Seo MD, Velamakanni S, Ishiyama N, Stathopoulos PB, Rossi AM, Khan SA *et al*. Structural and functional conservation of key domains in InsP3 and ryanodine receptors. *Nature* **2012**. PMID [22286060](#). doi:[10.1038/nature10751](#).
104. Cai X, Clapham DE. Ancestral Ca²⁺ signaling machinery in early animal and fungal evolution. *Mol Biol Evol* **2012**. PMID [21680871](#). doi:[10.1093/molbev/msr149](#).
105. Prole DL, Taylor CW. Identification of intracellular and plasma membrane calcium channel homologues in pathogenic parasites. *PLoS One* **2011**. PMID [22022573](#). doi:[10.1371/journal.pone.0026218](#).
106. Plattner H, Verkhratsky A. Ca²⁺ signalling early in evolution—all but primitive. *J Cell Sci* **2013**. PMID [23729741](#). doi:[10.1242/jcs.127449](#).
107. Plattner H. Molecular aspects of calcium signalling at the crossroads of unikont and bikont eukaryote evolution—the ciliated protozoan Paramecium in focus. *Cell Calcium* **2015**. PMID [25601027](#). doi:[10.1016/j.ceca.2014.12.002](#).
108. Patel S, Cai X. Evolution of acidic Ca²⁺ stores and their resident Ca²⁺-permeable channels. *Cell Calcium* **2015**. PMID [25591931](#). doi:[10.1016/j.ceca.2014.12.005](#).
109. Mackrill JJ. Ryanodine receptor calcium release channels: an evolutionary perspective. *Adv Exp Med Biol* **2012**. PMID [22453942](#). doi:[10.1007/978-94-007-2888-2_7](#).
110. Venkatesh K, Hasan G. Disruption of the IP3 receptor gene of Drosophila affects larval metamorphosis and ecdysone release. *Curr Biol* **1997**. PMID [9273145](#). doi:[10.1016/s0960-9822\(06\)00221-1](#).
111. Joshi R, Venkatesh K, Srinivas R, Nair S, Hasan G. Genetic dissection of itpr gene function reveals a vital requirement in aminergic cells of Drosophila larvae. *Genetics* **2004**. PMID [15020420](#). doi:[10.1534/genetics.166.1.225](#).
112. Banerjee S, Lee J, Venkatesh K, Wu CF, Hasan G. Loss of flight and associated neuronal rhythmicity in inositol 1,4,5-trisphosphate receptor mutants of Drosophila. *J Neurosci* **2004**. PMID [15356199](#). doi:[10.1523/jneurosci.0656-04.2004](#).
113. Srikanth S, Wang Z, Tu H, Nair S, Mathew MK, Hasan G *et al*. Functional properties of the Drosophila melanogaster inositol 1,4,5-trisphosphate receptor mutants. *Biophys J* **2004**. PMID [15189860](#). doi:[10.1529/biophysj.104.040121](#).

114. Kume S, Muto A, Aruga J, Nakagawa T, Michikawa T, Michikawa T *et al.* The Xenopus IP₃ receptor: structure, function, and localization in oocytes and eggs. *Cell* **1993**. PMID [8387895](#). doi:[10.1016/0092-8674\(93\)90142-d](#).
115. Matsumoto M, Nakagawa T, Inoue T, Nagata E, Tanaka K, Takano H *et al.* Ataxia and epileptic seizures in mice lacking type 1 inositol 1,4,5-trisphosphate receptor. *Nature* **1996**. PMID [8538767](#). doi:[10.1038/379168a0](#).
116. van de Leemput J, Chandran J, Knight MA, Holtzclaw LA, Scholz S, Cookson MR *et al.* Deletion at ITPR1 underlies ataxia in mice and spinocerebellar ataxia 15 in humans. *PLoS Genet* **2007**. PMID [17590087](#). doi:[10.1371/journal.pgen.0030108](#).
117. Futatsugi A, Nakamura T, Yamada MK, Ebisui E, Nakamura K, Uchida K *et al.* IP₃ receptor types 2 and 3 mediate exocrine secretion underlying energy metabolism. *Science* **2005**. PMID [16195467](#). doi:[10.1126/science.1114110](#).
118. Molitor A, Lederle A, Radosavljevic M, Sapuru V, Zavorka Thomas ME, Yang J *et al.* A pleiotropic recurrent dominant ITPR3 variant causes a complex multisystemic disease. *J Exp Med* **2024**. PMID [39270020](#). doi:[10.1084/jem.20232178](#).
119. Hara K, Shiga A, Nozaki H, Mitsui J, Takahashi Y, Ishiguro H *et al.* Total deletion and a missense mutation of ITPR1 in Japanese SCA15 families. *Neurology* **2008**. PMID [18579805](#). doi:[10.1212/01.wnl.0000311912.05593.1e](#).
120. Marelli C, van de Leemput J, Johnson JO, Tison F, Thauvin-Robinet C, Picard F *et al.* SCA15 due to large ITPR1 deletions in a cohort of 333 white families with dominant ataxia. *Arch Neurol* **2011**. PMID [21555639](#). doi:[10.1001/archneurol.2011.81](#).
121. Iwaki A, Kawano Y, Miura S, Shibata H, Matsuse D, Li W *et al.* Heterozygous deletion of ITPR1, but not SUMF1, in spinocerebellar ataxia type 16. *J Med Genet* **2008**. PMID [17932120](#). doi:[10.1136/jmg.2007.053942](#).
122. Novak MJ, Sweeney MG, Li A, Treacy C, Chandrashekar HS, Giunti P *et al.* An ITPR1 gene deletion causes spinocerebellar ataxia 15/16: a genetic, clinical and radiological description. *Mov Disord* **2010**. PMID [20669319](#). doi:[10.1002/mds.23223](#).
123. Durr A. Autosomal dominant cerebellar ataxias: polyglutamine expansions and beyond. *Lancet Neurol* **2010**. PMID [20723845](#). doi:[10.1016/s1474-4422\(10\)70183-6](#).
124. Huang L, Chardon JW, Carter MT, Friend KL, Dudding TE, Schwartzentruber J *et al.* Missense mutations in ITPR1 cause autosomal dominant congenital nonprogressive spinocerebellar ataxia. *Orphanet J Rare Dis* **2012**. PMID [22986007](#). doi:[10.1186/1750-1172-7-67](#).
125. Zambonin JL, Bellomo A, Ben-Pazi H, Everman DB, Frazer LM, Geraghty MT *et al.* Spinocerebellar ataxia type 29 due to mutations in ITPR1: a case series and review of this emerging congenital ataxia. *Orphanet J Rare Dis* **2017**. PMID [28659154](#). doi:[10.1186/s13023-017-0672-7](#).
126. Gerber S, Alzayady KJ, Burglen L, Bremond-Gignac D, Marchesin V, Roche O *et al.* Recessive and Dominant De Novo ITPR1 Mutations Cause Gillespie Syndrome. *Am J Hum Genet* **2016**. PMID [27108797](#). doi:[10.1016/j.ajhg.2016.03.004](#).
127. McEntagart M, Williamson KA, Rainger JK, Wheeler A, Seawright A, De Baere E *et al.* A Restricted Repertoire of De Novo Mutations in ITPR1 Cause Gillespie Syndrome with Evidence for Dominant-Negative Effect. *Am J Hum Genet* **2016**. PMID [27108798](#). doi:[10.1016/j.ajhg.2016.03.018](#).
128. Ronkko J, Molchanova S, Revah-Politi A, Pereira EM, Auranen M, Toppila J *et al.* Dominant mutations in ITPR3 cause Charcot-Marie-Tooth disease. *Ann Clin Transl Neurol* **2020**. PMID [32949214](#). doi:[10.1002/acn3.51151](#).
129. Cabello-Murgui J, Jimenez-Jimenez J, Vilchez JJ, Azorin I, Marti-Martinez P, Millet E *et al.* ITPR3-associated neuropathy: Report of a further family with adult onset intermediate

- Charcot-Marie-Tooth disease. *Eur J Neurol* **2024**. PMID [39287469](#). doi:[10.1111/ene.16466](#).
130. Beijer D, Dohrn MF, Rebelo A, Danzi MC, Grosz BR, Ellis M *et al.* A recurrent missense variant in ITPR3 causes demyelinating Charcot-Marie-Tooth with variable severity. *Brain* **2025**. PMID [38938188](#). doi:[10.1093/brain/awae178](#).
 131. Hytonen MK, Ronkko J, Hundi S, Jokinen TS, Suonto E, Teravainen E *et al.* IP3 receptor depletion in a spontaneous canine model of Charcot-Marie-Tooth disease 1J with amelogenesis imperfecta. *Dis Model Mech* **2025**. PMID [39804930](#). doi:[10.1242/dmm.052078](#).
 132. Ghosh TK, Eis PS, Mullaney JM, Ebert CL, Gill DL. Competitive, reversible, and potent antagonism of inositol 1,4,5-trisphosphate-activated calcium release by heparin. *J Biol Chem* **1988**. PMID [3136153](#). doi:[10.1016/s0021-9258\(18\)37923-7](#).
 133. Kobayashi S, Kitazawa T, Somlyo AV, Somlyo AP. Cytosolic heparin inhibits muscarinic and alpha-adrenergic Ca²⁺ release in smooth muscle. Physiological role of inositol 1,4,5-trisphosphate in pharmacomechanical coupling. *J Biol Chem* **1989**. PMID [2509451](#). doi:[10.1016/s0021-9258\(19\)84670-7](#).
 134. Peppiatt CM, Collins TJ, Mackenzie L, Conway SJ, Holmes AB, Bootman MD *et al.* 2-Aminoethoxydiphenyl borate (2-APB) antagonises inositol 1,4,5-trisphosphate-induced calcium release, inhibits calcium pumps and has a use-dependent and slowly reversible action on store-operated calcium entry channels. *Cell Calcium* **2003**. PMID [12767897](#). doi:[10.1016/s0143-4160\(03\)00026-5](#).
 135. Bootman MD, Collins TJ, Mackenzie L, Roderick HL, Berridge MJ, Peppiatt CM. 2-aminoethoxydiphenyl borate (2-APB) is a reliable blocker of store-operated Ca²⁺ entry but an inconsistent inhibitor of InsP3-induced Ca²⁺ release. *FASEB J* **2002**. PMID [12153982](#). doi:[10.1096/fj.02-0037rev](#).
 136. Gafni J, Munsch JA, Lam TH, Catlin MC, Costa LG, Molinski TF *et al.* Xestospongins: potent membrane permeable blockers of the inositol 1,4,5-trisphosphate receptor. *Neuron* **1997**. PMID [9331361](#). doi:[10.1016/s0896-6273\(00\)80384-0](#).
 137. Marchant JS, Beecroft MD, Riley AM, Jenkins DJ, Marwood RD, Taylor CW *et al.* Disaccharide polyphosphates based upon adenophostin A activate hepatic D-myo-inositol 1,4,5-trisphosphate receptors. *Biochemistry* **1997**. PMID [9335535](#). doi:[10.1021/bi971397v](#).